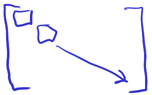
Algorithms and Operation Counting

ASEN 3502 — Computational Methods

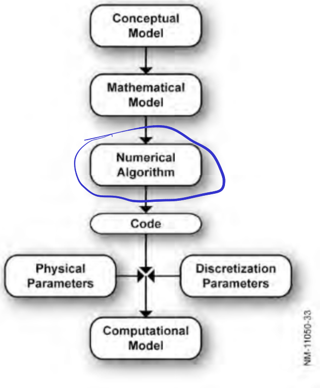
Please grab a handout in the back!

HW 1 - Due Thursday

Lab
min through 1.3
rec through 1.5



x = det



Today

- Algorithms
- Operation Counting
- Asymptotic Notation

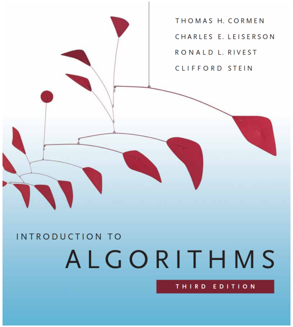
$O(n)$

ASME V&V 10, *Guide for Verification and Validation in Computational Solid Mechanics*.

What is an algorithm?

"Informally, an algorithm is any well-defined computational **procedure** that takes some value, or set of values, as **input** and produces some value, or set of values, as **output**."

— Cormen et al.



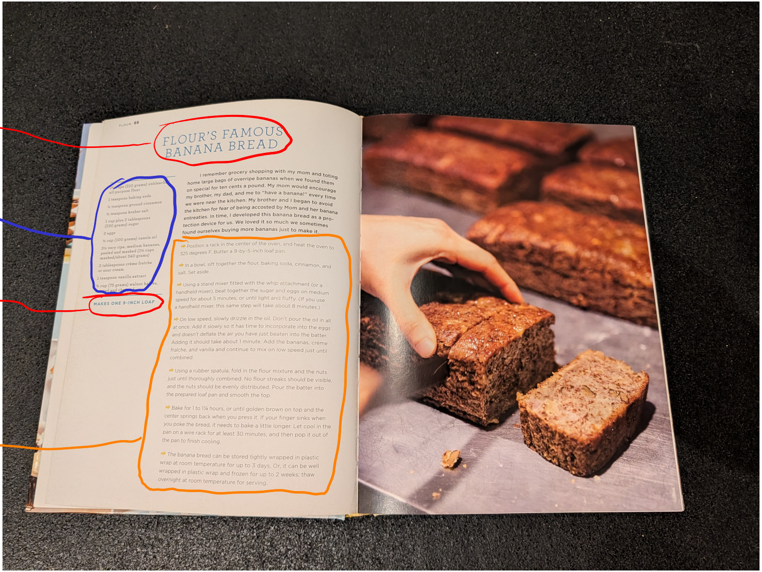
Introduction to Algorithms
Cormen, Leiserson, Rivest, and Stein

mitpress.mit.edu

Another type of algorithm

Outputs

Inputs



Procedure

Why algorithms?

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    > "An algorithm is said to be **correct** if, for every input instance, it halts with the correct output." — Cormen et al.

  - We can make precise and rigorous statements without having to deal with issues like roundoff error, memory allocation, etc.

How do we represent algorithms?

Matrix-vector multiplication: three ways

English

To get entry r of $\{b\}$, multiply each entry in row r of $[A]$ by the matching entry of $\{x\}$, and add up the results. Do this for every row.

Code

```
b = zeros(n,1);
for r = 1:n
    for c = 1:n
        b(r) = b(r) ...
             + A(r,c)*x(c);
    end
end
```

Pseudocode

input : $n \times n$ matrix $[A]$, vector $\{x\}$
output : $\{b\} = [A]\{x\}$

for $r \leftarrow 1$ **to** n **do**
    $b[r] \leftarrow 0$
    for $c \leftarrow 1$ **to** n **do**
        $b[r] \leftarrow b[r] + A[r,c] \cdot x[c]$
    end
end

"What separates pseudocode from "real" code is that in pseudocode, we employ whatever expressive method is most clear and concise to specify a given algorithm." — Cormen et al.

Analysis of algorithms

- *Analyzing* an algorithm means predicting the resources that an algorithm requires
  - For today, we will count floating point operations ("Flops"): multiplication, division, addition, and subtraction of floating point numbers

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- **Discuss with your neighbor:** Consider the equation $[A]\{x\} = \{b\}$. As n gets larger, which requires more floating point operations? Finding $\{x\}$ given $[A]$ and $\{b\}$, or finding $\{b\}$ given $[A]$ and $\{x\}$?

① $b = Ax$

② $Ax = b$ Harder if using Gaussian Elimination

Matrix-vector multiplication

input : $n \times n$ matrix $[A]$, vector $\{x\}$
output : $\{b\} = [A]\{x\}$

	flops	times
for $r \leftarrow 1$ **to** n **do**	0	$\sum_{r=1}^{n} 1 = n$
$b[r] \leftarrow 0$	0	n
for $c \leftarrow 1$ **to** n **do**	0	n^2
$b[r] \leftarrow b[r] + A[r,c] \cdot x[c]$	2	n^2
end		
end		

$$\sum_{r=1}^{n}\sum_{c=1}^{n} 1 = \sum_{r=1}^{n} n = n^2$$

FLOPS: $2n^2$

Gaussian elimination

input : $n \times n$ matrix $[A]$, right-hand side $\{b\}$
output : solution $\{x\}$ of $[A]\{x\} = \{b\}$

// forward elimination
for $c \leftarrow 1$ **to** $n-1$ **do**
  for $r \leftarrow c+1$ **to** n **do**
    $f \leftarrow A[r,c]\,/\,A[c,c]$
    for $j \leftarrow c$ **to** n **do**
      $A[r,j] \leftarrow A[r,j] - f \cdot A[c,j]$
    end
    $b[r] \leftarrow b[r] - f \cdot b[c]$
  end
end

// back substitution
for $r \leftarrow n, n-1, \ldots, 1$ **do**
  $s \leftarrow b[r]$
  for $j \leftarrow r+1$ **to** n **do**
    $s \leftarrow s - A[r,j] \cdot x[j]$
  end
  $x[r] \leftarrow s\,/\,A[r,r]$
end

flops	times
0	$\sum_{i=1}^{n-1} 1 = n-1$
0	$\frac{n^2-n}{2}$
1	
0	$\frac{n^3}{3}+O(n^2)$
2	
2	$\frac{n^2-n}{2}$
0	n
0	n
0	$\frac{n^2-n}{2}$
2	
1	n

Useful sum identities

$\sum_{i=1}^{m} c f(i) = c \sum_{i=1}^{m} f(i)$ | $\sum_{i=1}^{m} f(i)+g(i) = \sum_{i=1}^{m} f(i) + \sum_{i=1}^{m} g(i)$

$\sum_{i=1}^{m} 1 = m$ | $\sum_{i=k}^{m} 1 = m-k+1$

$\sum_{i=1}^{m} i = \frac{m(m+1)}{2}$ | $\boxed{\sum_{i=1}^{m} i^2 = \frac{m(m+1)(2m+1)}{6} = \frac{m^3}{3}+O(m^2)}$

$$\sum_{c=1}^{n-1}\sum_{r=c+1}^{n} 1 = \sum_{c=1}^{n-1} n-c\cancel{-1+1} = \sum_{c=1}^{n-1} n - \sum_{c=1}^{n-1} c = n(n-1) - \frac{(n-1)n}{2} = \frac{n^2-n}{2}$$

$$\sum_{c=1}^{n-1}\sum_{r=c+1}^{n}\sum_{j=c}^{n} 1 = \sum_{c=1}^{n-1}\sum_{r=c+1}^{n}(n-c+1) = \sum_{c=1}^{n-1}(n-c+1)(n-c+1) = \sum(-c+n+1)(-c+n+1)$$
$$= \sum_{c=1}^{n-1} c^2 - 2(n+1)\sum_{c=1}^{n-1} c + (n+1)^2\sum_{c=1}^{n-1} 1 = \frac{(n-1)^3}{3} + O(n^2) = \frac{n^3}{3}+O(n^2)$$

FLOPS: $\frac{n^2-n}{2} + 2\left(\frac{n^3}{3} + O(n^2)\right) + 2\frac{n^2-n}{2}$

FLOPS $\boxed{2\frac{n^3}{3}+O(n^2)} + 2\frac{n^2-n}{2} + n$

	flops	times
function $[C] = \mathrm{matmult}([A],[B])$		
for $i = 1$ **to** n	0	n
for $j = 1$ **to** n	0	n^2
$c_{ij} = 0$	0	$\vdots$
for $k = 1$ **to** n	0	n^3
$c_{ij} = c_{ij} \underline{+} \underline{a_{ik}\, b_{kj}}$	2	n^3
end		
end		
end		
return $[C]$		
end		

Asymptotic notation

- Asymptotic notation is a tool for keeping track of only the terms that are most important as a function approaches an asymptote

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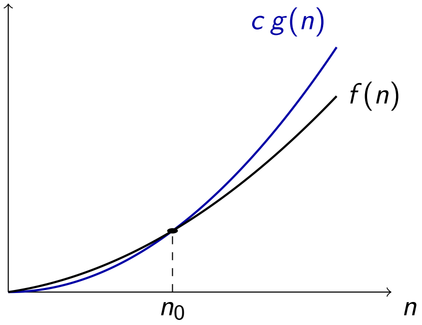
Definition

$f(n) = O(g(n))$ as $n \to \infty$ if there exist positive constants c and n_0 such that

$$0 \le f(n) \le c\, g(n) \quad \text{for all } n \ge n_0$$



$f(n) \le c\, g(n)$ for all $n \ge n_0$

$\leftarrow \quad f(n) = 3n^2 + 5n + 2$

$f(n) = 3n^2 + 5n + 2$

$c = 4.0$

$n_0 = 6$

then $0 \leq f(n) \leq c\,g(n)$ for all $n \geq n_0$

Therefore $f(n) = O(n^2)$

$c \geq$ sum of coefficients

$n_0 = 1$

$c = 10$

$n_0 = 1$

Useful Rules

1) $c f(n) = O(f(n)) \quad \forall\ c > 0$
2) if $f_1(n) = O(n^a)$ and $f_2(n) = O(n^b)$, $f_1(n) + f_2(n) = O(n^{\max(a,b)})$
3) if $f_1(n) = O(n^a)$ and $f_2(n) = O(n^b)$, $f_1(n) \cdot f_2(n) = O(n^{a+b})$
4) if $|f(n)|$ is bounded, $f(n) = O(1)$

Example: $f(n) = \underset{O(n^2)}{n^2} + \underbrace{12n}_{O(n)}\underbrace{\sin(n)}_{O(1)} + \underbrace{50}_{\substack{50 \cdot 1 \\ O(1)}} = O(n^2)$

Gaussian elimination

input : $n \times n$ matrix $[A]$, right-hand side $\{b\}$
output : solution $\{x\}$ of $[A]\{x\} = \{b\}$

	flops	Order of times called
`// forward elimination`		
for $c \leftarrow 1$ **to** $n-1$ **do** $\leq n$	×	$O(n)$
for $r \leftarrow c+1$ **to** n **do** $\leq n$	×	$O(n^2)$
$f \leftarrow A[r,c] \,/\, A[c,c]$	✓	
for $j \leftarrow c$ **to** n **do** $\leq n$	×	$O(n^3)$
$A[r,j] \leftarrow A[r,j] - f \cdot A[c,j]$	✓	
end	×	
$b[r] \leftarrow b[r] - f \cdot b[c]$	✓	$O(n^2)$
end		
end		
`// back substitution`		
for $r \leftarrow n, n-1, \ldots, 1$ **do**	×	$O(n)$
$s \leftarrow b[r]$	×	
for $j \leftarrow r+1$ **to** n **do**	×	$O(n^2)$
$s \leftarrow s - A[r,j] \cdot x[j]$	✓	
end		
$x[r] \leftarrow s \,/\, A[r,r]$	✓	$O(n)$
end		

$O(n^3)$ Flops